Vector Integration: Line Integral of a Vector Field

1. Line Integral

Name of Concept

Line Integral of a Vector Field (Work Done by a Force Field).

Core Mathematical Formulas

The differential position vector $d\bar{r}$ represents an infinitesimal displacement along a path:

$$d\bar{r} = dx\bar{i} + dy\bar{j} + dz\bar{k}$$

The line integral of a vector field $\bar{F}$ along a curve C is given by:

$$\int_C \bar{F} \cdot d\bar{r} = \int_C (f_1\,dx + f_2\,dy + f_3\,dz)$$

Beginner-Friendly Explanation of Operations

1. **What is a Line Integral?** Instead of integrating a function over a straight intervals on the x-axis (like standard single-variable calculus), a line integral integrates a vector field along a specific curved path C in 3D space.
2. **The Vector Field ($\bar{F}$):** Think of $\bar{F}$ as a collection of force vectors distributed through space, such as wind currents or gravitational pull. It is broken into its spatial components: $f_1\bar{i} + f_2\bar{j} + f_3\bar{k}$.
3. **The Differential Displacement ($d\bar{r}$):** As you walk along the curve C, every microscopic step you take can be broken down into tiny steps along the coordinate axes: a tiny shift in x (dx), a tiny shift in y (dy), and a tiny shift in z (dz).
4. **The Dot Product Core ($\bar{F} \cdot d\bar{r}$):** At every single point along the path, we compute the dot product between the force vector $\bar{F}$ acting at that location and your tiny step vector $d\bar{r}$. Mechanically, you multiply corresponding components together:

$$\bar{F} \cdot d\bar{r} = (f_1 \cdot dx) + (f_2 \cdot dy) + (f_3 \cdot dz)$$

> This dot product measures how much of the force field is pushing *with* you or *against* you in the direction of your movement. If the force is perpendicular to your step, the dot product is 0. In physical terms, this calculates the microscopic work done by the field during that tiny step.

5. **The Integration Symbol ($\int_C$):** The integral symbol accumulates all these microscopic work values from the starting point of the curve C to its terminal point, yielding the total scalar value of work done along the path.